

APPLICATION NOTE

Return-aware validation of an earnings coefficient under household masking in the Current Population Survey

Hyun Hak Kim*

ABSTRACT

Earnings missing for entire households can remove logistic event support, while replacing earnings can shift the coefficient targeted by an analysis. We study the association between main-job earnings and multiple jobholding in three Current Population Survey arrays, using 30,300 controlled household masks. We combine focal likelihood checks, diagnostic method references, and separate reporting of reference availability, interval return and coverage. The earnings association changes with adjustment for usual hours. At 25% household retention, ordinary complete-case intervals return on 68.4–89.0% of masks; extended focal intervals return throughout. In synthetic data, conditional-mean substitution covers the common target on only 0–0.4% of repetitions even when its basis contains the true conditional mean. Adding a missing quadratic term in nonlinear conditions improves coverage by 38.4–50.8 percentage points, but coverage remains 59.9–71.7%. Resampled panels distinguish failure of a finite full-vector reference from focal nonidentification and show why return and coverage comparisons need matched denominators. These findings separate support, substitution and learner problems. Conclusions concern selected released arrays under known experimental masking, not actual CPS nonresponse or population coverage.

KEYWORDS

Current Population Survey; estimand alignment; logistic separation; missing covariates; clustered masking; simulation validation

1. Introduction

Weekly earnings are widely used as a covariate in labour-market analysis, yet an earnings coefficient can become difficult to estimate when earnings are unavailable for part of a clustered sample. Two problems then arise together. Removing incomplete records can leave too little event support for a finite multivariable logistic fit. Filling in or integrating over unavailable earnings introduces additional choices about the conditional model and its use in the analysis. Even an accurate conditional mean need not preserve a nonlinear estimating equation when substituted for the missing covariate. An interval that is reported only after a successful fit can therefore look reliable while concealing either numerical nonreturn or displacement from the intended complete-data coefficient.

We examine this problem using the January 2025, July 2025 and January 2026 Basic Monthly Current Population Survey (CPS). The application relates released usual weekly earnings to multiple jobholding among selected employed wage and salary workers in outgoing rotation groups. The three monthly arrays contain 7,085, 6,390 and 6,344 people in 4,801, 4,254 and 4,332 households. Multiple jobholding occurs for 5.6–5.8% of the selected people. Event support is much thinner within some adjustment

* Professor, Department of Economics, Kookmin University, Seoul, Korea

groups: for example, only 9, 16 and 16 households contain a multiple jobholder with less than high-school education in the three arrays. Earnings allocation flags occur for 30.6–32.9% of otherwise eligible people, and hundreds of households contain both allocated and unallocated records. These observed patterns motivate the analysis, while their structure does not identify a household-level response law.

The substantive question is how the association between main-job earnings and multiple jobholding depends on adjustment for main-job hours, and how reliably that specified association can be recovered after earnings are hidden. Earnings opportunities and work schedules both enter explanations of multiple jobholding (Lalé, 2015). Weekly earnings combine pay rates and hours, so an hours-adjusted projection and one omitting hours answer different descriptive questions. Neither identifies why a worker takes a second job.

The study therefore uses the released records with unallocated earnings as fixed complete validation arrays and imposes a known experimental law. A household is retained with probability p , and the earnings regressor is hidden for every person in a nonretained household; outcomes, adjustment variables and auxiliary predictors remain available. The complete-data quantity is the earnings coefficient in an unpenalized logistic projection for each selected monthly array. It is an association defined by the released records. The experiment does not estimate actual CPS response probabilities, survey-design population coverage or a causal effect.

The first data challenge is loss of support after household masking. Logistic likelihoods can have recession directions even when optimization software returns a large finite vector (Geyer, 2009). A nuisance coefficient may diverge while a specified linear contrast remains identified on the limiting likelihood face. For the earnings coefficient, we therefore distinguish failure of the full coefficient vector from failure of the focal contrast. The implemented extended complete-case procedure checks the whole recession cone, tests whether the earnings contrast belongs to the row space of the remaining design, and fits the resulting finite problem. Firth logistic regression provides a secondary finite comparator, with its penalized center kept distinct from the unpenalized target (Kosmidis and Firth, 2021).

The second challenge is estimand alignment under learned earnings. Standard missing-covariate scores can use a conditional mean, integrate over a fitted distribution, or augment an observed-data score (Robins et al., 1994). Their uncertainty must also account for the estimated imputation quantities (Robins and Wang, 2000). In a finite validation array, both the replacement rule and a misspecified learner can make the probability-averaged score solve for a method-specific projection. The need to align an imputation model with the substantive analysis is also central to Bartlett et al. (2015); our deterministic score comparisons do not evaluate their multiple-imputation procedure. We calculate a diagnostic reference with the all-data learner held fixed and compare its coverage with coverage of the complete-data coefficient. This is not an exact decomposition of bias and variance for a learner refitted under each mask.

The application combines these ideas in a return-aware validation workflow. Using complete validation records, we first certify the reference projection, then check focal estimability under masking and record every planned evaluation. Reference availability, point and interval return, joint return-and-coverage, and coverage conditional on return have separate denominators. This ordering matters because conditional coverage can exclude exactly the masks on which support is lost. Reporting method failure is an established requirement in method comparisons (Wünsch et al., 2025). Our application combines that reporting with a check of whether an earnings contrast survives nuisance separation and with paired diagnostics of the replacement rule and learner.

With genuinely unobserved earnings, an analyst can define the desired projection and audit observed-data geometry, but cannot calculate its unknown complete-data value or its realized coverage. The complete-record experiments assess the procedures before such deployment.

Three validation structures examine different parts of the problem. First, six procedures are applied to 30,300 shared household masks across the three fixed monthly arrays. At $p = 0.25$, ordinary complete-case intervals return on 89.0%, 79.8% and 68.4% of the masks; their joint coverage is consequently well below their coverage conditional on return. The extended focal procedure returns on all fixed-array evaluations, including 2,107 masks without a certified full-vector complete-case root. Second, a paired synthetic intervention holds the complete data, target and mask fixed while adding a prespecified quadratic term to the earnings learner. It tests whether an observed error pattern is responsive to a specific learner correction. Third, household-resampled CPS panels vary the underlying array at $H = 1, 200$ and $4, 800$ and retain ten masks within each of 200 panel IDs. This experiment distinguishes loss of a finite full-vector reference from nonreturn of a fitted procedure. Fixing complete data is generally insufficient for assessing missing-data methods across new samples (Morris et al., 2024). Here it defines a conditional mask-law question; the synthetic and resampled arrays examine separate questions about changes to that array.

The application also reveals target sensitivity before any value is hidden. Under the main adjustment, the complete-data earnings coefficients are -0.102 , -0.123 and -0.003 in the three months. Removing usual-hours terms changes them to -0.271 , -0.249 and -0.156 . The masking results must therefore be interpreted together with the definition of the monthly projection. A method may return an interval even when its estimating equation targets a coefficient different from the prespecified projection.

Section 2 defines the records, targets, controlled laws and performance denominators. Section 3 presents the analysis workflow and interval constructions. Section 4 gives the three-month CPS application, and Section 5 reports the paired learner and household-panel experiments. Section 6 concludes with an analyst checklist. The supplement gives the restricted focal-estimability result, complete cell tables, computational rules and reproducibility specifications.

2. CPS data challenges, targets and validation design

2.1. Selected records and two challenges

We use the January 2025, July 2025 and January 2026 Basic Monthly CPS public-use files (U.S. Census Bureau, 2025b,c, 2026b). Within each month, we select employed civilians aged 16 or older in outgoing rotation groups 4 or 8 whose main job is in government or private wage and salary employment. We require $\text{PEMJOT} \in \{1, 2\}$ and $\text{PXMJOT} = 0$, nonnegative released weekly earnings and $\text{PRWERNAL} = 0$. The outcome is $y_i = 1$ for $\text{PEMJOT} = 1$ (multiple jobs in the reference week) and zero for code 2. Released earnings are $E_i = \text{PTERNWA}_i / 100$ dollars and the focal regressor is $z_i = \log(1 + E_i)$, following the public-use record layouts (U.S. Census Bureau, 2025a, 2026a). The released values retain CPS rounding and disclosure treatment (U.S. Census Bureau, 2023). Earnings refer to usual weekly pay at the main job, including usual overtime, commissions and tips, rather than the sum over all jobs (U.S. Bureau of Labor Statistics, 2026). The hours variable PEHRUSL1 also describes usual main-job hours. Thus the earnings and hours measures describe usual activity, whereas the outcome concerns

Table 1. Selected CPS arrays and support diagnostics. Allocation rates use otherwise eligible records before the allocation exclusion. The final column counts households containing a multiple jobholder with less than high-school education.

Month	People	HH	Events	Event (%)	Alloc. (%)	Low-educ. event HH
January 2025	7,085	4,801	408	5.759%	31.796%	9
July 2025	6,390	4,254	355	5.556%	32.899%	16
January 2026	6,344	4,332	355	5.596%	30.560%	16

the survey reference week.

The first challenge is support. A household mask can remove every event from an adjustment pattern and destroy a finite full-vector logistic fit, while the limiting likelihood may still identify the earnings contrast. A returned focal point estimate also needs a separately validated standard error. The second challenge is alignment. Procedures that predict or integrate over hidden z_i can solve an estimating equation centered at a method-specific projection through the replacement rule or learner misspecification. Numerical return and the reference covered by the interval must therefore be recorded separately.

The allocation patterns in Table 1 motivate these challenges but do not define the experiment’s observation law. An allocation flag need not mean complete earnings nonresponse, and 370–442 otherwise eligible households have mixed flags. We use the selected released arrays as fixed validation data and impose controlled household masks. “Complete data” refers only to these released arrays.

The two January arrays share 1,306 household keys and 1,763 person keys; 1,740 person keys also pass a basic sex and age-consistency check. This is not a verified longitudinal linkage. We report the months separately and do not treat them as independent external replications. The masks are generated separately within each fixed monthly array.

2.2. Targets and controlled validation structures

Let $x_i \in \mathbb{R}^{24}$ span age and usual-hours spline terms, sex, marital-status, race and class-of-worker indicators, and an education-code term with two education thresholds. Supplement Section S5.1 specifies all columns and their transformations. Put $w_i = (1, z_i, x_i^\top)^\top$. The primary target for each month is the earnings component β_F of the finite logistic projection

$$\frac{1}{N} \sum_{i=1}^N w_i \{y_i - \text{expit}(w_i^\top \theta_F)\} = 0, \quad \theta_F = (\alpha_F, \beta_F, \gamma_F^\top)^\top. \quad (1)$$

This selected-array association is defined without assuming that the logistic conditional mean is correct. Prespecified sensitivity targets remove the six hours terms or weight the score by normalized PWORWGT. They change the projection and are not measures of masking bias.

In a fixed-array experiment, independent $R_h \sim \text{Bernoulli}(p)$ determine whether earnings are visible for household h ; all other analysis and learner variables remain visible. Six procedures share each mask. The January 2025 cells at $p = 0.25$ and 0.50 use 6,400 archived masks each, and the other seven month- p cells use 2,500

each, giving 30,300 masks. A paired synthetic experiment holds 1,000 complete arrays, targets and masks fixed in each of eight conditions while changing one learner term. A third experiment resamples January 2025 donor households to create 200 panels at each of $H = 1,200$ and 4,800, then applies ten masks at $p = 0.25$ and 0.50. Resampled panel IDs, with their ten masks kept together, are the Monte Carlo units. The panel reference is available only when the prespecified complete-data *full-vector* root is certified. Failure of this rule does not imply that the earnings contrast is unidentified on a limiting likelihood face.

2.3. Return-aware performance measures

For planned unit r and procedure j , let T_r indicate that the prespecified finite full-vector reference was certified, A_{rj} a returned interval and H_{rj} coverage of its stated reference. We report

$$r_j^{\text{int}} = \frac{\sum_r A_{rj}}{|\mathcal{R}|}, \quad c_j^{\text{joint}} = \frac{\sum_r T_r A_{rj} H_{rj}}{\sum_r T_r}, \quad c_j^{\text{cond}} = \frac{\sum_r T_r A_{rj} H_{rj}}{\sum_r T_r A_{rj}}. \quad (2)$$

Thus nonreturn remains in the planned return denominator and counts as zero in joint coverage when $T_r = 1$. Conditional coverage requires both $T_r = 1$ and a returned interval. To compare return and coverage on the same set, we also report

$$r_j^{\text{int}|T} = \frac{\sum_r T_r A_{rj}}{\sum_r T_r}, \quad c_j^{\text{joint}} = r_j^{\text{int}|T} c_j^{\text{cond}}.$$

The identity uses return conditional on reference availability, not overall return r_j^{int} . Bias and RMSE require a certified reference and point estimate; length comparisons require both intervals. Fixed-array rates receive exact binomial Monte Carlo intervals, while continuous and paired summaries use 4,000 paired bootstrap draws. Panel bootstraps resample the 200 panel IDs. For panels with a certified reference and all ten point estimates and intervals returned, we also compare mean reported SE^2 with within-panel mask variance and an untruncated between-panel component. Complete definitions and denominator rules are in the supplement.

3. A return-aware and estimand-aligned approach

3.1. Scores and analysis sequence

In the validation experiment, we certify the complete-data projection and store it only for evaluation. Each candidate procedure then uses the always-observed variables and visible earnings, with observed-data geometry checked for CC and Extended CC. Write $R_i = R_{h(i)}$, where $h(i)$ is the household of person i , and let $s_i(z; \theta) = w_i(z) \{y_i - \text{expit}(w_i(z)^\top \theta)\}$. Five-fold household cross-fitting estimates the conditional mean of $z_i = \log(1 + E_i)$ by ridge regression. The CPS learner includes the outcome, demographics, job characteristics, geography and response mode. Training uses only visible earnings outside the evaluation fold. The variance $\hat{v}_i$ is constructed from training residual second moments and within-household residual products, with fixed variance floors; it is not a ridge variance regression. The full specification is in

$$t_{Mi} = s_i(\hat{\mu}_i; \theta), \quad t_{Gi} = \sum_{l=1}^{30} \omega_l s_i(\hat{\mu}_i + \sqrt{2\hat{v}_i} \xi_l; \theta), \quad (3)$$

where ξ_l are Gauss–Hermite nodes and the weights are normalized to $\sum_l \omega_l = 1$. The four full-vector equations are

$$\begin{aligned} \Psi_{CC} &= \sum_i R_i s_i, & \Psi_j &= \sum_i \{R_i s_i + (1 - R_i) t_{ji}\}, \quad j = M, G, \\ \Psi_A &= \sum_i \{t_{Gi} + R_i(s_i - t_{Gi})/p\}. \end{aligned} \quad (4)$$

These define complete case, conditional mean, Gaussian integration and the augmented score. The learner, penalty grid, variance floors and folds are fixed before the performance results are inspected.

3.2. Focal estimability and interval return

For the observed logistic design W_R , set $S = \text{diag}(2y_R - 1)W_R$ and $\mathcal{C}_R = \{v : Sv \geq 0\}$. The implemented extended CC procedure finds the union of rows perfectly fitted along directions in $\mathcal{C}_R$ and lets $W_{\mathcal{F}_R}$ denote the remaining design. The earnings contrast c is returned only if it is invariant along the entire recession cone and $c \in \text{row}(W_{\mathcal{F}_R})$. A finite reduced logistic problem then estimates the contrast. This check can preserve focal inference when a nuisance coefficient diverges; it does not by itself validate the interval. Proposition S1.2 in the supplementary material gives a mask-law limit under bounded-household, regular-bulk and sparse-event assumptions. That proposition concerns a restricted profile model and supplies no general coverage guarantee for the CPS design or the fitted-learner procedures.

For a returned full-vector or reduced fit, the household-mask variance has the common form

$$\hat{s}_j^2 = \frac{H}{H-1} (1-p) \sum_{h: R_h=1} q_{jh}^2, \quad (5)$$

Here H is the total number of households before masking. For a full-vector fit, let $A_j = -N^{-1} \partial \Psi_j / \partial \theta^\top$, holding the fitted learner fixed, and $\ell_j = A_j^{-\top} c$, with $c = (0, 1, 0, \dots, 0)^\top$. Then $q_{CC,h} = N^{-1} \ell_{CC}^\top \sum_{i \in h} s_i$ and $q_{A,h} = (Np)^{-1} \ell_A^\top \sum_{i \in h} (s_i - t_{Gi})$, evaluated at the corresponding fitted coefficient. Mean and Gaussian additionally use derivatives of the refitted learner, holding the selected ridge penalty fixed and weighting the generated-score sensitivity by $1 - p$. Supplement Sections S5.3–S5.4 give these derivatives and the reduced-fit and penalty-aware definitions of q_{jh} . Each interval is $\hat{\beta}_j \pm 1.959964 \hat{s}_j$. It is withheld after rank, root, derivative or positive-SE failure. Firth logistic regression is a secondary comparator, using a certified Jeffreys-penalized fit and a penalty-aware household-mask linearization. Its center remains distinct from the unpenalized target (Kosmidis and Firth, 2021).

Table 2. Complete-data earnings coefficients and descriptive transforms. The odds ratio is $\exp\{\beta_F \log(1001/501)\}$. Probability change averages fitted $P(y = 1 \mid E = 1000, x_i) - P(y = 1 \mid E = 500, x_i)$ over the selected records, using normalized outgoing weights for the weighted projection and equal weights otherwise.

Month	Projection	$\hat{\beta}_F$	OR _{500→1000}	Probability change (pp)
January 2025	Main adjustment	-0.102	0.932	-0.38
	Hours excluded	-0.271	0.829	-1.10
	Outgoing weights	-0.074	0.950	-0.26
July 2025	Main adjustment	-0.123	0.918	-0.46
	Hours excluded	-0.249	0.842	-0.97
	Outgoing weights	-0.058	0.960	-0.20
January 2026	Main adjustment	-0.003	0.998	-0.01
	Hours excluded	-0.156	0.898	-0.60
	Outgoing weights	-0.018	0.988	-0.06

3.3. Evaluation references and scope

CC, Augmented, Extended CC and Firth are evaluated against β_F . Mean and Gaussian also receive an evaluation-only reference obtained by fitting their learner with all earnings visible and solving the corresponding probability-averaged score. This diagnoses a shift in the estimating equation; the reference is neither supplied to a masked fit nor equated with the expectation of an estimator whose learner is refitted under each mask. Residual error about that reference can include finite-sample bias as well as variance error. Root acceptance follows the rank, score, Newton-correction and start-comparison rules specified in the supplement. No failed mask is redrawn and no penalty or adjustment term is changed in response to its result. Table grids, learner derivatives, numerical tolerances and failure codes are reported in the supplement.

4. Application to earnings and multiple jobholding

4.1. Target sensitivity

Table 2 shows that the selected-array association changes before masking. The main earnings coefficients are -0.102 , -0.123 and -0.003 . Removing hours makes every coefficient more negative, while outgoing weights move the two 2025 estimates toward zero. Each row is a different projection, so these differences are target sensitivity rather than method bias.

For interpretation, doubling main-job weekly earnings from \$500 to \$1,000 corresponds to a fitted probability change of -0.38 , -0.46 and -0.01 percentage points under the main adjustment, compared with -1.10 , -0.97 and -0.60 when hours are omitted. With hours held fixed, the projection compares earnings at a given main-job schedule; without hours, variation associated with that schedule also enters the earnings relationship. The near-zero main-adjustment coefficient in January 2026 therefore does not describe a near-zero association under every adjustment question. These comparisons concern selected released records. They establish neither a causal response to higher earnings nor a statistically significant change across months. They identify which descriptive coefficient the masking analysis is being asked to recover.

Table 3. Interval return / joint coverage under fixed-array masking (%).

Month	p	Masks	CC	Mean	Gaussian	Augmented	Extended CC	Firth
January 2025	.25	6,400	89.0/84.7	100/88.4	100/93.5	100/95.5	100/95.0	100/94.3
	.50	6,400	99.8/94.3	100/86.3	100/93.8	100/94.7	100/94.5	100/94.4
	.75	2,500	100/94.9	100/89.5	100/94.8	100/95.0	100/94.9	100/94.4
July 2025	.25	2,500	79.8/72.4	100/92.4	100/89.2	100/91.6	100/90.8	100/89.6
	.50	2,500	99.8/91.8	100/88.4	100/92.2	100/93.1	100/92.0	100/91.6
	.75	2,500	100/93.8	100/90.9	100/94.1	100/94.2	100/93.8	100/93.4
January 2026	.25	2,500	68.4/65.1	100/93.0	100/93.6	100/94.4	100/95.0	100/94.2
	.50	2,500	96.4/90.4	100/92.9	100/93.9	100/94.4	100/93.8	100/93.6
	.75	2,500	100/94.3	100/94.2	100/94.7	100/94.7	100/94.4	100/94.0

4.2. Return, focal estimability and joint coverage

At $p = 0.25$, CC returns an interval on 89.0%, 79.8% and 68.4% of the three monthly masks. Conditional coverage is 95.2%, 90.7% and 95.1%, but joint coverage falls to 84.7%, 72.4% and 65.1%. The gap is produced by nonreturn, not by reclassifying failed fits. CC return rises above 96% at $p = 0.50$ and is essentially complete at $p = 0.75$.

Extended CC returns on all 30,300 masks, including all 2,107 masks without a certified CC interval. In 15 prespecified January 2025 failures, the removed rows exactly match the sparse groups identified by the separation certificate and the earnings contrast passes the row-space check. For example, one mask removes 112 Asian-only rows; Extended CC estimates -0.330 with interval $[-0.570, -0.091]$. These cases demonstrate focal availability, while Extended CC joint coverage of 90.8–95.0% shows that interval performance remains a separate empirical question.

4.3. Paired and reference diagnostics

On shared masks, Augmented minus Extended CC joint-coverage differences are positive in eight of nine cells, but eight pointwise bootstrap intervals include zero. Augmented intervals are 2.55–6.03% shorter in seven cells; the other two comparisons are unresolved. This supports an efficiency gain in several audited conditions without a uniform ranking.

Mean returns everywhere but its method reference differs from β_F by -0.189 to -0.002 . Common-target coverage is 86.3–94.2%, compared with 92.2–95.9% for its displaced reference. Gaussian reference displacement is smaller, between -0.0064 and -0.0012 , although common-target coverage is still 89.2–94.8%. Firth also returns everywhere, but its penalty center and mask-linearization interval differ from Extended CC. Complete paired intervals and reference diagnostics appear in the supplement.

5. Validation for similarly structured data

5.1. Paired learner intervention

Each of 1,000 repetitions contains 800 two-person households, with 20% in a rare-event stratum and $m \in \{2, 12\}$ event households there. The generated earnings-like variable is

$$z = 0.4x + 0.8y + 0.2G + \delta(x^2 - 1) + 0.6u + 0.8v, \quad (6)$$

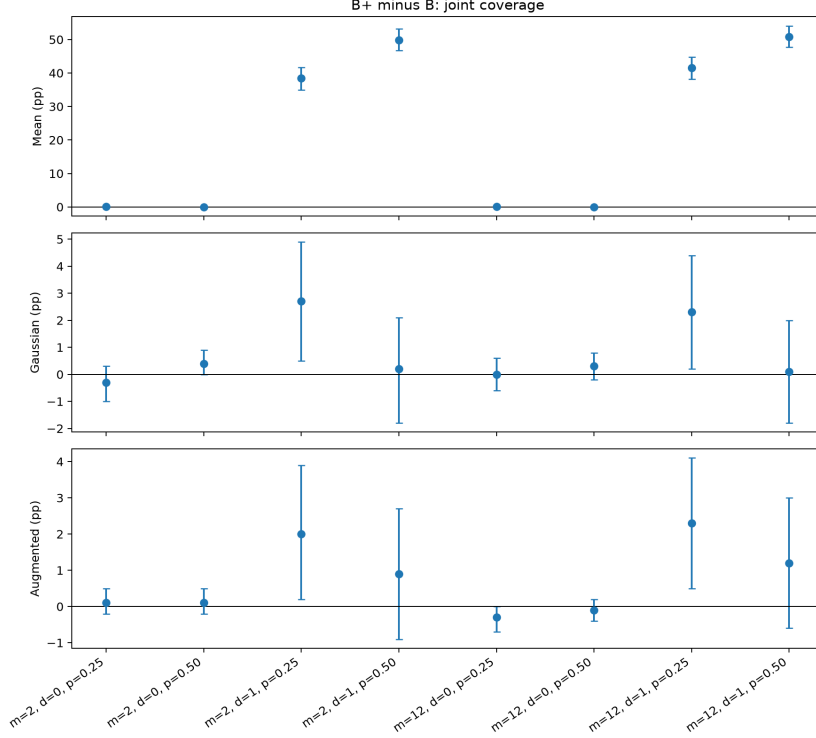


Figure 1. Paired $B^+ - B$ change in joint coverage. Bars are 95% bootstrap Monte Carlo intervals; each condition contains 1,000 pairs.

where u and v are independent standard normal variables, with u shared by the two people in a household. Here G indicates the rare stratum; x is marginally standard normal with within-household correlation 0.25. Bulk outcomes have probability $\text{expit}(-1 + 0.3x)$; in the rare stratum, one person in each of m uniformly selected households is an event and everyone else is a nonevent. The complete-data target is the unpenalized logistic projection on $(1, z, x, G)$, computed separately for each generated array. Supplement Section S3.1 gives the construction. Eight conditions cross m , $\delta \in \{0, 1\}$ and $p \in \{0.25, 0.50\}$. The paired intervention changes $B = (1, x, G, y)$ to $B^+ = (1, x, G, y, x^2 - 1)$ while holding the complete array, target and mask fixed.

Figure 1 reports changes; Table 4 shows why the absolute levels also matter. When $\delta = 0$, the basis contains the true conditional mean, but Mean joint coverage is only 0–0.4% under either basis. Coverage of its diagnostic method reference is 93.1–94.7%, with mean reference displacement of 0.878–2.781. Conditional mean accuracy alone is insufficient because, for the nonlinear logistic score, $s_i\{E(z_i | B_i); \theta\}$ generally differs from $E\{s_i(z_i; \theta) | B_i\}$. Estimation and ridge shrinkage can add further discrepancies. The small $B^+ - B$ changes at $\delta = 0$ therefore do not indicate satisfactory coverage.

When $\delta = 1$, the added term raises Mean joint coverage by 38.4–50.8 points and reduces its reference shift from 0.271–1.121 to 0.044–0.076. Common-target coverage nevertheless remains 59.9–71.7%, whereas method-reference coverage is 93.2–94.1%. Gaussian and Augmented gains are 2.0–2.7 points at $p = 0.25$ and no more than 1.2 points at $p = 0.50$. Return is unchanged in every cell. This paired intervention isolates the effect of adding a known missing mean term. It does not eliminate the nonlinear substitution problem or establish a complete decomposition of remaining bias and interval error.

Table 4. Mean joint coverage (%) before and after learner enrichment. Every cell contains 1,000 repetitions and has 100% interval return.

m	p	$\delta = 0$		$\delta = 1$	
		B	B^+	B	B^+
2	0.25	0.3	0.4	33.3	71.7
2	0.50	0.0	0.0	10.3	60.2
12	0.25	0.1	0.2	28.9	70.4
12	0.50	0.0	0.0	9.1	59.9

Table 5. Return and joint coverage in $H = 1,200$ panels. Overall return uses 2,000 evaluations; reference-conditional return and joint coverage use 1,740. The bracketed 95% Monte Carlo interval resamples 200 panel IDs. The empirical bootstrap has no variation for the all-return cells.

p	Method	Overall (%)	Given reference (%)	Joint (%)
0.25	Extended CC	100.00	100.00	90.69
	Augmented	86.70	99.60 [99.19, 99.94]	95.92
0.50	Extended CC	100.00	100.00	92.13
	Augmented	87.00	100.00	93.74

5.2. Household-resampled CPS panels

The prespecified finite full-vector reference is certified in 174 of 200 $H = 1,200$ panels and every $H = 4,800$ panel. In all 26 excluded small panels, the original design has rank 26 but separation removes rows from the limiting likelihood. The earnings contrast passes the recession-cone and row-space checks in every such panel. A subsequent diagnostic fit also certifies a finite extended complete-data earnings coefficient in all 26 (Supplement Section S4.1). These additional fits do not replace the prespecified reference or its coverage denominator.

In the smaller panels, overall CC interval return is 3.9% at $p = 0.25$ and 37.6% at $p = 0.50$. Restricted to certified-reference panels, return is 4.48% and 43.16%; conditional coverage of 94.9% and 92.9% then gives joint coverage of 4.3% and 40.1%. Extended CC returns in every small-panel evaluation. Augmented returns on 86.7–87.0% overall, but on 99.60% and 100% of evaluations with a certified reference (Table 5). On this latter set its joint coverage exceeds Extended CC by 5.23 and 1.61 points. The separate comparison of pairs with both intervals returned gives longer Augmented intervals. Most of the overall return difference arises in the 26 excluded panels: Extended CC returns in all 260 evaluations at each p there, while Augmented returns once at $p = 0.25$ and never at $p = 0.50$. Those intervals are outside the prespecified common-target coverage comparison.

At $H = 4,800$, both procedures return throughout. Their joint-coverage differences are -0.15 and 0.15 points, with bootstrap intervals containing zero, and Augmented intervals are shorter by 0.0139 and 0.0150. On balanced ten-mask panels, mean reported SE^2 divided by total empirical error variance ranges from 0.920 to 1.008 for both procedures in the large panels. At $H = 1,200$, $p = 0.25$, it is 3.441 for Augmented and 0.636 for Extended CC. The decomposition is conditional on complete return and retains negative between-panel estimates without truncation. These results distinguish small-panel full-vector support problems from learner error and large-panel interval length differences.

6. Conclusion

The application yields three conclusions. The selected-array earnings association changes across month, adjustment and weighting. Conditional coverage among successful fits can conceal serious nonreturn after sparse household support is lost. The replacement rule, learner specification and array support create distinct problems. A mean model containing the true mean can still give severe undercoverage when its predictions are substituted into a nonlinear score. Adding a missing term helps in the nonlinear conditions but does not resolve all displacement. Smaller resampled panels can lose a finite full-vector reference while retaining an estimable earnings contrast.

Augmented and Extended CC are complementary. Augmentation can shorten intervals when support is adequate; Extended CC can retain an estimable earnings contrast when nuisance components diverge. Their return, coverage and length rankings change with the validation structure, so the experiments do not establish uniform dominance.

For validation with complete records, analysts should define the projection and sensitivity targets before masking, audit household event support and focal estimability, and keep reference availability distinct from interval return. Coverage and return comparisons should use a common eligible set, with all planned evaluations also reported. Both the learner and underlying household array need stress tests. For actual incomplete records, the unknown complete-data coefficient and its coverage cannot be certified; the applicable steps are target definition, observed-data estimability checks, prespecified fitting and justified uncertainty and sensitivity analysis. The conclusions concern a known experimental mask law on released CPS arrays. Actual CPS non-response, survey-design population inference and causal earnings effects require additional identification and variance analyses.

Data availability statement

The Basic Monthly CPS public-use files are available from the U.S. Census Bureau (U.S. Census Bureau, 2025b,c, 2026b). The accompanying replication package contains the exact input releases, analysis code, per-fit outputs and instructions for reproducing the tables, figures and diagnostic calculations. Supplement Sections S3–S5 specify the designs and computational procedures. Data-version hashes identify the releases used in this study.

References

- Bartlett, J. W., Seaman, S. R., White, I. R., and Carpenter, J. R. (2015). Multiple imputation of covariates by fully conditional specification: Accommodating the substantive model. *Statistical Methods in Medical Research*, 24(4):462–487.
- Geyer, C. J. (2009). Likelihood inference in exponential families and directions of recession. *Electronic Journal of Statistics*, 3:259–289.
- Kosmidis, I. and Firth, D. (2021). Jeffreys-prior penalty, finiteness and shrinkage in binomial-response generalized linear models. *Biometrika*, 108(1):71–82.
- Lalé, E. (2015). Multiple jobholding over the past two decades. *Monthly Labor Review*. U.S. Bureau of Labor Statistics. <https://doi.org/10.21916/mlr.2015.7>.
- Morris, T. P., White, I. R., Cro, S., Bartlett, J. W., Carpenter, J. R., and Pham, T. M. (2024). Comment on Oberman & Vink: Should we fix or simulate the complete data in simulation studies evaluating missing data methods? *Biometrical Journal*, 66(1):2300085.
- Robins, J. M., Rotnitzky, A., and Zhao, L. P. (1994). Estimation of regression coefficients when

- some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866.
- Robins, J. M. and Wang, N. (2000). Inference for imputation estimators. *Biometrika*, 87(1):113–124.
- U.S. Bureau of Labor Statistics (2026). Frequently asked questions about earnings data from the Current Population Survey. Technical documentation, accessed 19 September 2026. <https://www.bls.gov/cps/earnings-faqs.htm>.
- U.S. Census Bureau (2023). 2023 Current Population Survey user note. Technical documentation, February 8.
- U.S. Census Bureau (2025a). 2025 basic CPS public use record layout plus industry and occupation code list. Technical documentation.
- U.S. Census Bureau (2025b). Basic monthly Current Population Survey: January 2025. Public-use data file, jan25pub.zip.
- U.S. Census Bureau (2025c). Basic monthly Current Population Survey: July 2025. Public-use data file, jul25pub.zip.
- U.S. Census Bureau (2026a). 2026 basic CPS public use record layout plus industry and occupation code list. Technical documentation.
- U.S. Census Bureau (2026b). Basic monthly Current Population Survey: January 2026. Public-use data file, jan26pub.zip.
- Wünsch, M., Herrmann, M., Noltenius, E., Mohr, M., Morris, T. P., and Boulesteix, A.-L. (2025). Rethinking the handling of method failure in comparison studies. *Statistics in Medicine*, 44(23–24):e70257.